

The Category Theory of Cymatic Yogurt Computing: A Thermodynamically-Aware Resource Theory

Flyxion

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Abstract

Cymatic Yogurt Computing (CYC) is a speculative but mathematically concrete bio-stochastic computing paradigm in which vibrated microbial cultures serve as computational media. Patterns on yogurt cultures, shaped by cymatic excitation and transferred through sieving operations, embody information processing under explicit thermodynamic constraints. In this essay we develop a categorical and resource-theoretic formalism for CYC. We define a category of culture states, equip it with a symmetric monoidal and comonoidal structure, introduce entropy and Fisher information as functorial scalar invariants, and show how closed monoidal structure enables higher-order “control cultures” that implement learning-like adaptation. The resulting framework clarifies how bio-stochastic substrates can be rigorously modeled as thermodynamically-aware resource theories and connects naturally to broader work on scalar–vector field theories and semantic infrastructure.

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1 Introduction: The Need for Categorical Rigor in Bio-Stochastic Systems

1.1 The computational crisis and the digital bottleneck

Classical computing is dominated by the von Neumann architecture: a clean separation between memory, processing, and control, coordinated through a global clock and a narrow instruction pipeline. This architecture has been astonishingly successful, but it carries structural liabilities that become acute for highly parallel, noisy, and energy-constrained regimes.

First, the *digital bottleneck*: even if the underlying hardware contains vast numbers of degrees of freedom (transistors, photonic modes, molecular states), the usable interface is compressed into a serial stream of discrete instructions. Moving information across memory hierarchies and interconnects often costs more energy and time than the logical operations themselves. Second, the von Neumann model is largely indifferent to thermodynamic structure. Landauer’s principle is acknowledged in theory but rarely appears as a first-class design constraint; energy is something to minimize, not a quantity to be structurally modeled.

These limitations become visible in neuromorphic hardware, synthetic biology, and other emerging domains where computation is massively parallel, inherently stochastic, and embedded in non-equilibrium thermodynamic flows. In such contexts, the usual abstractions—

bit-level logic, Turing machines, RAM models—are not so much wrong as insufficiently structured: they do not track which transformations are physically cheap, which require irreversible mixing, and which depend on fine-grained spatial coherence.

1.2 Introducing Cymatic Yogurt Computing (CYC)

Cymatic Yogurt Computing (CYC) pushes this situation to an intentionally extreme and playful limit. Consider a shallow tray of yogurt seeded with microbial cultures and subjected to controlled vibrational patterns. The resulting standing-wave fields, diffusion gradients, and microbial responses produce rich spatial patterns: nodal lines, clustering domains, and traveling waves. These patterns are treated as *computational states*.

Computation proceeds in three coupled steps:

1. **Cymatic pattern formation:** vibrational modes and boundary conditions induce spatial modes in the culture, shaping density and metabolic fields.
2. **Sieving and transfer:** the patterned culture is passed through a physical sieve or stencil, producing a filtered pattern that is embedded onto a fresh culture.
3. **Entropic descent and reorganization:** the newly seeded culture relaxes toward thermodynamic equilibrium, generating further structure along the way.

The core mechanism is thus “computation via vibration, pattern transfer, and entropic descent.” The substrate is biological and stochastic; the state space is defined over spatial domains; and the cost of computation is unavoidably thermodynamic.

1.3 Thesis: CYC as a thermodynamically-aware resource theory

The central claim of this essay is that Category Theory furnishes the appropriate structural language for CYC. We will show that:

- Culture states and their transformations form a well-defined category $\mathcal{C}$ whose objects encode spatially distributed patterns as sheaves over a domain, and whose morphisms model sieve–transfer operations.
- $\mathcal{C}$ admits a symmetric monoidal structure $(\mathcal{C}, \otimes, I)$ corresponding to parallel juxtaposition of culture bins, with entropy acting as a monoidal functor that tracks thermodynamic cost.
- Duplication and erasure operations endow objects with comonoid structure, making CYC a *resource theory*: objects are resources, morphisms are allowed processes, and comonoid axioms express the laws of resource budgeting under entropic constraints.

- A closed monoidal upgrade equips $\mathcal{C}$ with internal homs $[C, D]$ that represent “higher-order yogurt functions,” or control cultures that implement morphisms between other cultures. This supplies a structural basis for learning and adaptation in CYC.

In short, CYC becomes a thermodynamically-aware resource theory whose categorical structure is rich enough to support higher-order computation, yet grounded in physical operations on microbial media.

2 Objects and Foundations: The Category of Culture States $\mathcal{C}$

2.1 Culture states as computational sheaves

We begin by specifying the objects of our category.

Definition 1 (Culture state). *A culture state C is a pair $C = (X, \mathcal{F})$ where*

- $X \subset \mathbb{R}^2$ *is a compact spatial domain representing the footprint of the culture tray;*
- $\mathcal{F} \in \mathbf{Sh}(X)$ *is a pattern sheaf assigning to each open set $U \subseteq X$ a space $\mathcal{F}(U)$ of admissible local configurations (e.g. microbial density, nutrient and metabolite concentrations), together with restriction maps satisfying the usual sheaf axioms.*

We write $\text{Ob}(\mathcal{C})$ for the class of all such culture states.

The sheaf $\mathcal{F}$ encodes local-to-global consistency: patterns overlapping in space must agree on intersections. This is crucial for modeling diffusion, reaction–diffusion dynamics, and morphogenetic constraints. Intuitively, $\mathcal{F}$ describes the *computable pattern space* over the culture domain.

2.2 Entropy and Fisher information as scalar functionals

CYC is explicitly thermodynamic, so each culture state carries global entropy and information measures.

Definition 2 (Entropy functional). *Let $C = (X, \mathcal{F})$ be a culture state, and let $\sigma_C : X \rightarrow \mathbb{R}_{\geq 0}$ denote a local entropy density associated with a particular realization of the sheaf data. The thermodynamic entropy of C is*

$$S(C) = \int_X \sigma_C(\mathbf{x}) \, d\mathbf{x}. \quad (1)$$

In parallel, we introduce a dual measure of structure.

Definition 3 (Fisher information). *Given a parametric family of local state distributions $\{p_\theta(\cdot \mid \mathbf{x})\}$ over $\mathcal{F}(\{\mathbf{x}\})$, we define the local Fisher information as*

$$I_C(\mathbf{x}) = \mathbb{E} \left[\left(\frac{\partial}{\partial \theta} \log p_\theta(\cdot \mid \mathbf{x}) \right)^2 \right], \quad (2)$$

and the global Fisher information as

$$I(C) = \int_X I_C(\mathbf{x}) d\mathbf{x}. \quad (3)$$

Heuristically, $S(C)$ quantifies thermodynamic “spread” while $I(C)$ quantifies structural precision. CYC computation can be viewed as a structured compromise: patterns evolve under operations that, on average, increase S while selectively concentrating I into computationally meaningful modes.

3 Morphisms: The Core Sieving Step

3.1 Sieve transfer as a composite morphism

Morphisms in $\mathcal{C}$ represent physically admissible operations on culture states. The core step in CYC is the *sieve transfer*: patterned yogurt is forced through a stencil onto a fresh culture.

Definition 4 (Sieve transfer). *Given culture states $C_s = (X_s, \mathcal{F}_s)$ and $C_t = (X_t, \mathcal{F}_t)$, a sieve transfer is a morphism $f : C_s \rightarrow C_t$ of the form*

$$f = \psi \circ \phi, \quad (4)$$

where ϕ is a filtration operator on the source pattern, and ψ is a propagation operator embedding the filtered pattern into the target culture.

We now specify these components more explicitly.

3.2 Filtration: coarse-graining as semantic compression

The filtration operator ϕ models the effect of the physical sieve or stencil that selectively transmits features of the source pattern.

Definition 5 (Filtration operator). *Let $K : \mathbb{R}^2 \rightarrow \mathbb{R}_{\geq 0}$ be a sieve kernel encoding the geometry and pore structure of the sieve (or more general non-linear weight). A filtration operator $\phi : C_s \rightarrow C'_s$ acts at the level of sections by a non-linear convolution*

$$\phi(\mathcal{F}_s)(\mathbf{x}) = \Phi \left(\int_{X_s} K(\mathbf{x} - \mathbf{y}) \mathcal{F}_s(\mathbf{y}) d\mathbf{y} \right), \quad (5)$$

for a suitable non-linearity Φ reflecting thresholding, saturation, or other biological responses.

Physically, ϕ implements semantic compression: fine-grained variations below the sieve scale are suppressed; only features that couple strongly to K survive. From the perspective of information theory, ϕ is typically non-invertible and strictly reduces Fisher information:

$$I(C'_s) \leq I(C_s), \quad (6)$$

with strict inequality in generic cases.

3.3 Propagation: stochastic embedding and entropy gradients

Once filtered, the pattern is projected onto a fresh culture.

Definition 6 (Propagation operator). *Let $C'_s = (X_s, \mathcal{F}'_s)$ be the filtered source and $C_t = (X_t, \mathcal{F}_t)$ a fresh target. A propagation operator $\psi : C'_s \rightarrow C_t$ is a stochastic monomorphism determined by a transition kernel*

$$P(\mathbf{z} \mid \mathbf{x}) : X_s \times X_t \rightarrow [0, 1], \quad (7)$$

such that sections of $\mathcal{F}_t$ are drawn from $\mathcal{F}'_s$ according to P . Formally, for suitable test functionals G ,

$$\mathbb{E}[G(\mathcal{F}_t)] = \int_{X_t} \int_{X_s} G(\mathcal{F}'_s(\mathbf{x})) P(\mathbf{z} \mid \mathbf{x}) d\mathbf{x} d\mathbf{z}. \quad (8)$$

The act of landing the filtered culture onto the fresh medium typically introduces an immediate local entropy gradient ∇S on X_t , driving further entropic descent and pattern formation. In this sense, ψ is both a structural embedding and an entropic kick, aligning with the idea that computation in CYC is always embedded in a thermodynamic story.

3.4 The identity morphism as a zero-work baseline

The identity morphism in $\mathcal{C}$ corresponds to a vanishingly idealized sieve transfer that performs no net work on the pattern.

Definition 7 (Identity morphism). *For a culture state $C = (X, \mathcal{F})$, the identity morphism $\text{id}_C : C \rightarrow C$ is realized as the limit in which*

- the filtration kernel K approaches a Dirac delta ($K \rightarrow \delta$), so that ϕ performs no coarse-graining; and
- the propagation kernel $P(\mathbf{z} \mid \mathbf{x})$ becomes a deterministic identity map with zero stochastic variance and no net entropy gradient ($\nabla S = 0$).

Physically this limit is unattainable: any real sieve introduces both irreversible loss of information and some thermodynamic cost. However, id_C serves as a conceptual baseline. The “distance” between a real sieve transfer f and id_C (e.g. in terms of changes in S and I) quantifies the entropic and informational work done by f .

4 Monoidal Structure: Parallelism and Resource Management

4.1 The monoidal product as parallel juxtaposition

We now introduce parallel composition.

Definition 8 (Monoidal product). *Given culture states $C_1 = (X_1, \mathcal{F}_1)$ and $C_2 = (X_2, \mathcal{F}_2)$ with disjoint domains, their monoidal product is*

$$C_1 \otimes C_2 = (X_1 \sqcup X_2, \mathcal{F}_1 \sqcup \mathcal{F}_2), \quad (9)$$

where $\sqcup$ denotes disjoint union of spaces and sheaves.

Morphisms tensor similarly: given $f_i : C_i \rightarrow D_i$, the monoidal product $f_1 \otimes f_2 : C_1 \otimes C_2 \rightarrow D_1 \otimes D_2$ acts componentwise. The associativity and symmetry constraints are provided by the canonical homeomorphisms between iterated disjoint unions, making $(\mathcal{C}, \otimes, I)$ a symmetric monoidal category once the unit object I is specified.

4.2 Entropy additivity for parallel cultures

Because the domains are disjoint, entropy behaves additively under the monoidal product.

Proposition 1 (Additivity of entropy). *Let $C_1 = (X_1, \mathcal{F}_1)$ and $C_2 = (X_2, \mathcal{F}_2)$ be culture states with disjoint domains. Then*

$$S(C_1 \otimes C_2) = S(C_1) + S(C_2). \quad (10)$$

Proof. By definition,

$$S(C_1 \otimes C_2) = \int_{X_1 \sqcup X_2} \sigma(\mathbf{x}) \, d\mathbf{x} = \int_{X_1} \sigma_1(\mathbf{x}) \, d\mathbf{x} + \int_{X_2} \sigma_2(\mathbf{x}) \, d\mathbf{x}, \quad (11)$$

which is exactly $S(C_1) + S(C_2)$. $\square$

This strict additivity has two important implications. First, parallelism incurs no “mysterious” entropic overhead purely by juxtaposing culture bins: the cost of running n computations in parallel is n times the cost of running one, at least in the idealized model where bins

do not interact. Second, it justifies modeling CYC as a resource theory in which resources combine under a strictly monoidal law for entropy.

4.3 The unit object as waste sink and null culture

We now introduce the neutral element for $\otimes$.

Definition 9 (Unit object). *The unit object I is either*

- *the empty culture $(\emptyset, \mathcal{F}_\emptyset)$, or*
- *a maximally homogenized culture $C_{\max}$ in which all spatial gradients have relaxed and the pattern sheaf carries only uniform sections.*

In both cases, I satisfies

$$C \otimes I \cong C \quad \text{and} \quad I \otimes C \cong C \tag{12}$$

for all C .

Operationally, it is convenient to treat I as a *waste sink*: a reservoir into which structured patterns can be discarded and homogenized.

4.4 Comonoid structure: duplication and erasure

To complete the resource-theoretic picture, we endow each culture state with a comonoid structure that models duplication and erasure.

Definition 10 (Comonoid on a culture state). *A culture state C is a comonoid in $(\mathcal{C}, \otimes, I)$ if it is equipped with morphisms*

$$\Delta : C \rightarrow C \otimes C \quad \text{and} \quad \epsilon : C \rightarrow I \tag{13}$$

satisfying the usual coassociativity and counitality axioms:

$$(\Delta \otimes \text{id}_C) \circ \Delta = (\text{id}_C \otimes \Delta) \circ \Delta, \tag{14}$$

$$(\epsilon \otimes \text{id}_C) \circ \Delta = \text{id}_C = (\text{id}_C \otimes \epsilon) \circ \Delta. \tag{15}$$

Duplication via physical splitting

The comultiplication $\Delta : C \rightarrow C \otimes C$ models splitting a single culture into two boxes and propagating the pattern into both halves. Physically, this might involve pouring a patterned culture into two receiving trays and allowing diffusion and growth to fill each.

The key thermodynamic insight is that duplication is not free. Even if the pattern is replicated with high fidelity, mixing and re-equilibration in each half generate additional entropy.

Theorem 1 (Thermodynamic cost of duplication). *Let C be a culture state with entropy $S(C)$. Then the duplicated state $\Delta(C)$ satisfies*

$$S(\Delta(C)) = 2S(C) + \Delta S_{\text{mix}}, \quad (16)$$

where $\Delta S_{\text{mix}} \geq 0$ is a mixing entropy term arising from the reorganization of the medium during splitting and propagation.

In particular, $\Delta S_{\text{mix}} > 0$ in generic cases, so duplication is thermodynamically irreversible. This mirrors no-cloning constraints in other physical theories: perfect duplication without entropy increase is forbidden.

Erasure as homogenization or autoclaving

The counit $\epsilon : C \rightarrow I$ represents erasure or consumption of a culture state.

Definition 11 (Counit as erasure). *The counit $\epsilon : C \rightarrow I$ is realized physically by operations such as*

- *homogenization: vigorous stirring or mixing that eliminates all spatial gradients and pattern structure; or*
- *autoclaving/thermal shock: driving the culture to a thermodynamic fixed point in which microbial life ceases and only uniform disorder remains.*

At the level of sheaves, ϵ collapses all sections of $\mathcal{F}$ to the unique section of $\mathcal{F}_\emptyset$, effectively forgetting all pattern information.

Erasure is, by construction, a maximally irreversible operation. The associated entropy production approximates the Landauer cost of destroying information:

$$S_{\text{dissipated}} \approx S_{\text{max}} - S(C), \quad (17)$$

where S_{max} is the entropy of the fully homogenized state.

Counitality ensures that duplication followed by erasure of one copy restores the original resource (up to approximations set by ΔS_{mix}): splitting a culture into two and immediately destroying one half leaves the other half essentially unchanged at the level of computational state.

5 Higher-Order Computation: Closed Monoidal Structure

5.1 Why internal homs matter

So far, $\mathcal{C}$ can express sequences and parallel combinations of sieve transfers. To capture *higher-order* behavior—cultures that encode and modify other cultures’ dynamics—we require an internal representation of morphisms. This is precisely what a closed monoidal structure provides.

Definition 12 (Closed monoidal category). *A symmetric monoidal category $(\mathcal{C}, \otimes, I)$ is (right) closed if for every pair of objects C, D there exists an object $[C, D]$ (the internal hom) and a natural isomorphism*

$$\mathrm{Hom}(A \otimes C, D) \cong \mathrm{Hom}(A, [C, D]) \quad (18)$$

that is natural in A .

In CYC terms, $[C, D]$ should be a *control culture* whose internal pattern encodes an entire family of sieve transfers from C to D .

5.2 Internal hom as control culture

We now propose an interpretation of $[C, D]$ for CYC.

Definition 13 (Control culture). *Given culture states $C_1 = (X_1, \mathcal{F}_1)$ and $C_2 = (X_2, \mathcal{F}_2)$, a control culture $[C_1, C_2]$ is a culture state $(X_{\mathrm{ctrl}}, \mathcal{F}_{\mathrm{ctrl}})$ equipped with a sheaf morphism*

$$\mathcal{F}_{\mathrm{ctrl}} \rightarrow \underline{\mathrm{Hom}}(\mathcal{F}_1, \mathcal{F}_2), \quad (19)$$

where $\underline{\mathrm{Hom}}(\mathcal{F}_1, \mathcal{F}_2)$ is the Hom-sheaf assigning to each open set $U \subseteq X_1$ the set of sheaf morphisms $\mathcal{F}_1|_U \rightarrow \mathcal{F}_2|_{\phi(U)}$ for suitable embeddings ϕ .

Physically, the control culture consists of specialized regulatory bacteria and chemical gradients. When brought into interaction with an input culture C_1 , the combined system

$$C_1 \otimes [C_1, C_2]$$

relaxes, through the usual CYC dynamics, into a pattern that matches the desired output state in C_2 . In other words, the control culture biases the filtration and propagation operators toward a specific family of morphisms $C_1 \rightarrow C_2$.

5.3 Currying, uncurrying, and learning in CYC

The closed structure asserts that giving a morphism

$$f : A \otimes C \rightarrow D$$

is equivalent to giving a morphism

$$\tilde{f} : A \rightarrow [C, D].$$

In the CYC picture, the first is a direct, “hard-wired” sieve pipeline that takes an auxiliary culture A and an input culture C and produces an output D . The second is a process by which the auxiliary culture A —perhaps a training dataset, environment, or “curriculum culture”—shapes the control culture $[C, D]$ itself.

This is precisely the categorical shape of learning:

- *Uncurried view*: apply a fixed control culture to many inputs in parallel to compute outputs.
- *Curried view*: treat observed input–output behavior as a morphism from an environment A into the space of possible control cultures.

In more concrete terms, a control culture could be evolved by repeatedly subjecting it to a training protocol: co-culturing with C , applying sieves and vibrational patterns, and selecting for those descendants that move the joint system toward the desired state in D . Over time, this implements a physical approximation to a map $A \rightarrow [C, D]$.

6 Conclusion: CYC as a Thermodynamically-Aware Resource Theory

6.1 Synthesis of the categorical structure

We can now summarize the categorical formalism of CYC:

- The category $\mathcal{C}$ of culture states has objects $C = (X, \mathcal{F})$ representing spatially distributed microbial patterns, and morphisms $f : C_s \rightarrow C_t$ representing sieve transfers composed from filtration and propagation operators.
- Entropy $S(C)$ and Fisher information $I(C)$ are scalar functionals on objects that quantify thermodynamic spread and structural precision, respectively.

- A symmetric monoidal structure $(\mathcal{C}, \otimes, I)$ models parallel juxtaposition of cultures, with entropy strictly additive under $\otimes$, ensuring that parallel computation is transparently accounted for in thermodynamic terms.
- Comonoid structure (Δ, ϵ) on objects captures duplication (with irreducible mixing entropy) and erasure (with Landauer-like dissipation), making CYC a resource theory in which objects are computational resources and morphisms are allowed physical processes.
- A closed monoidal upgrade introduces internal homs $[C, D]$ that represent control cultures encoding families of morphisms $C \rightarrow D$, supplying a natural setting for higher-order computation and learning.

6.2 Positioning within a broader theoretical ecosystem

While Cymatic Yogurt Computing is deliberately whimsical in its choice of substrate, the formal structure developed here is fully general. Any computational medium with the following properties

1. spatially extended states modeled by sheaves or related local-to-global data structures,
2. inherently stochastic and thermodynamically constrained transformations, and
3. physically meaningful duplication and erasure operations

can, in principle, be organized into a similar thermodynamically-aware resource theory.

In this sense, CYC can be seen as a toy model of more ambitious frameworks in which scalar–vector field theories, semantic infrastructures, and entropic alignment mechanisms are unified. The category $\mathcal{C}$ is a local caricature of much larger plenum-like systems where information and energy flow through structured, self-organizing media.

6.3 Future directions

Several extensions suggest themselves:

- **Enrichment over stochastic kernels.** One can refine the formalism by treating $\mathcal{C}$ as enriched over a category of stochastic matrices or Markov kernels, so that morphisms carry explicit probability structures and compositional noise models.
- **Compact closure and feedback.** Introducing duals and exploring compact closed or traced monoidal structure would allow feedback loops and cyclic sieve pipelines to be modeled explicitly, capturing the dynamics of resonant or recurrent CYC setups.

- **Higher sheaf-theoretic structure.** Replacing simple sheaves with more sophisticated objects (e.g. cosheaves or factorization algebras) could provide richer handles on locality, causality, and multi-scale organization in bio-stochastic computation.
- **Experimental toy models.** Finally, building small physical prototypes—using dye patterns, colloids, or simple microbial cultures under vibration and sieving—would test which parts of the categorical story survive contact with experimental reality.

Cymatic Yogurt Computing thus plays a dual role: it is both a playful thought experiment and a serious probe into what it means to treat computation as an explicitly thermodynamic, spatially extended, and categorically organized process. In situating CYC as a thermodynamically-aware resource theory, we obtain a template for future work on non-digital architectures—from yogurt cultures to cosmological plenums—that compute by falling, mixing, and reorganizing in the direction of entropy.